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Advancing Infiltration Modeling with Deep Learning: A Superior Alternative to Empirical Approaches in the Mitidja Plain

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***Abstract*— Irrigation planning and flood risk assessment are critical challenges, particularly in semi-arid regions. This study evaluates the performance of deep learning models—Long Short-Term Memory (LSTM) and Multi-Layer Perceptron (MLP)—in predicting infiltration rates across 70 experimental sites in the Mitidja Plain, Algeria, and compares them with traditional empirical models, namely Philips and SCS. Given the limitations of empirical models in capturing the complexity of infiltration dynamics, machine learning offers a powerful alternative for improving prediction accuracy.**

The LSTM model, designed with two layers (100 and 50 units) and dropout regularization, effectively captures temporal dependencies, while the MLP model, tested with different dimensional configurations, serves as a strong benchmark for static regression tasks. Results indicate that MLP outperforms all other models, achieving a coefficient of determination (R^2) of 0.95, compared to 0.81 for LSTM. In contrast, empirical models demonstrated lower accuracy, with SCS performing better than Philip. These results confirm that deep learning, particularly MLP, provides a more accurate and robust solution for infiltration modeling compared to classical empirical approaches.

These findings highlight the transformative potential of artificial intelligence in hydrological studies, offering enhanced precision for soil and water conservation strategies in semi-arid environments. This study contributes to optimizing infiltration estimation and supports the integration of advanced computational methods into hydrology for improved water resource management in the Mitidja Plain, Algeria. Future research should explore hybrid approaches that combine empirical knowledge with deep learning techniques for further refinement.

Keywords*— *Artificial Intelligence, Deep Learning, Hydrology, Infiltration Modeling

I. INTRODUCTION

Infiltration—the process through which water penetrates the soil surface—is a key component in hydrological modeling, directly affecting flood forecasting, drought resilience, and irrigation planning [1]. Factors like soil texture, antecedent moisture, and land cover significantly influence infiltration behavior. Coarse-textured soils typically allow higher infiltration, while fine or compacted soils restrict it. Vegetation also plays a critical role in enhancing or limiting infiltration depending on coverage density [2]. However, direct infiltration measurements in the field remain labor-intensive, expensive, and spatially limited. This reinforces the importance of predictive models in representing dynamic infiltration across broader spatial scales [3].

Traditionally, empirical models such as Philips and SCS have been widely applied to estimate infiltration rates. While easy to calibrate and interpret, these models are based on simplifying assumptions and often fall short in reflecting the nonlinearity and variability inherent in real-world infiltration processes [4]. In contrast, machine learning methods—particularly deep learning algorithms like Long Short-Term Memory (LSTM) and Multi-Layer Perceptron (MLP)offer data-driven alternatives capable of capturing complex temporal and nonlinear relationships in infiltration datasets. These models use soil properties and dynamic inputs to improve predictive performance [5]. This study focuses on the Mitidja Plain in Algeria, a critical agricultural zone experiencing rising water management pressures due to climate variability and unsustainable irrigation practices [6]. Seventy infiltration tests were conducted using a double ring infiltrometer to build a robust dataset across various soil textures. The primary objective is to compare the performance of deep learning models (LSTM and MLP) with empirical models (Philips and SCS) using multiple performance indicators including R^2, RMSE, NSE, and KGE. Results emphasize the superior accuracy of MLP in modeling infiltration, highlighting the value of integrating AI in modern hydrological practice and sustainable water management strategies for semi-arid regions like Mitidja [7].

II. MATERIALS AND METHODS

A. Study area

The Mitidja Plain, situated in northern Algeria, covers an area of approximately 1,700 km², with 120,000 to 130,000 hectares dedicated to agriculture. The region experiences a Mediterranean climate, characterized by hot summers with average temperatures around 33°C and mild, rainy winters averaging 6°C. Annual precipitation varies between 500 and 700 mm. Several major rivers, including the Oued Legaia, Hamiz, El Harrach, and Mazafran, traverse the plain, contributing to its hydrological dynamics. The area is known for its highly fertile alluvial and colluvial soils, which provide optimal conditions for agricultural activities [8].

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B. Infiltration Rate Measurement and Soil Sampling

The Infiltration rates were assessed using a double-ring infiltrometer, a widely used method for evaluating soil water absorption capacity. The device consists of two concentric metal rings inserted 40 mm into the soil, with both rings filled with water. The water level in the inner ring was recorded at one-minute intervals until the infiltration rate reached equilibrium. This approach provides valuable insights into soil infiltration behavior, essential for effective water resource management in agricultural landscapes.

Additionally, soil samples were collected from 70 experimental sites at a depth of 50 cm. These samples were prepared for laboratory analysis and classified based on their texture using a textural triangle, allowing for a comprehensive assessment of soil properties influencing infiltration dynamics. Fig.1 represents Location of the study area within the Mitidja plain, Algeria, and the experimental site used for infiltration measurements.

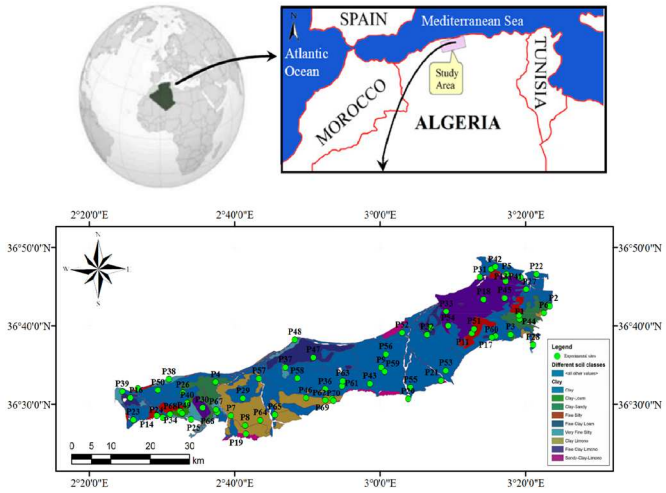


Fig .1. Geographical location and soil texture of the study area.

C. Machine learning and empirical models

1) Multi-Layer Perceptron (MLP)

The Multi-Layer Perceptron (MLP) is a deep learning model capable of capturing complex, nonlinear relationships in data. It consists of multiple layers of interconnected nodes, with weights adjusted during training using gradient descent to minimize prediction errors. In hydrology, MLP effectively models variables like rainfall, slope, and soil type [9]. Activation functions (e.g., ReLU, sigmoid) introduce nonlinearity, making MLP suitable for predicting infiltration by processing multidimensional data with strong generalization ability [10].

2) Long Short-Term Memory (LSTM)

LSTM is a type of recurrent neural network (RNN) designed to model time-dependent processes [11]. In this study, LSTM was used to predict infiltration rates using 1-minute interval data, incorporating inputs like time, soil moisture, and texture. Its memory cells and gating mechanisms allow it to learn long-term dependencies. Trained with the Adam optimizer and evaluated using metrics like NSE, KGE, CC, and RMSE, the LSTM model showed good performance in capturing the dynamic and nonlinear nature of infiltration.

3) Philip Model

Philip (1957a) proposed a model based on a semi-analytical solution of the Richards flow equation. In this solution,[12] used a time series to solve the Richards equation. Neglecting the higher order terms, the results of a two-term infiltration equation were derived in which both parameters have physical significance

$$f(t) = \frac{1}{2} S\, t^{-0.5} + K \qquad (1)$$

Where f(t) is the infiltration rate (mm/h) at time t, S is the Sorptivity parameter that is function of soil matrix forces (mm/h-0.5), t is time (h) and K is the soil parameter related to transmission of water through the soil or gravity force.

4) SCS model

The SCS model is an empirically developed approach to the water infiltration process [13], as follows:

$$I = \mathrm{a}t^{b} + 0.6985 \qquad (2)$$

Where, a and b are the equation constants. I is the cumulative infiltration rate (L).

D. Performance comparing parameters

Root Mean Square Error (RMSE), coefficient of correlation (r), The Nash-Sutcliffe Efficiency (NSE) and The Percent Bias (PBIAS), are widely adopted metrics for assessing the quality of predictions and facilitate the comparison of models across different scales [14].

$$RMSE = \sqrt{\frac{1}{N}\left(\sum_{i=0}^{n}(a_i - b_i)^2\right)} \qquad (3)$$

$$CC = \frac{\sum_{i=0}^{n}(ab) - \sum_{i=0}^{n}(a)\sum_{i=0}^{n}(b)}{\sqrt{\sum_{i=0}^{n}(a^2)(-(\sum_{i*=0}^{n} a)^2}\sqrt{\sum_{i=0}^{n}(b^2) - (\sum_{i*=0}^{n} b)^2}} \qquad (4)$$

$$NSE = 1 - \frac{\sum_{i=0}^{n}((b_i - a_i)^2)}{\sum_{i=0}^{n}\left((b_i - \underline{b})^2\right)} \qquad (5)$$

$$PBIAS = 100\frac{1}{n}\sum_{i=0}^{n}(a_i - b_i) \qquad (6)$$

where ai is the simulated value; bi is the observed value of infiltration rate; and n is the number of observations.

E. Methodology

1) Empirical Models

Two empirical models, Philip's two-term and the Soil Conservation Service (SCS), were calibrated in RStudio using nonlinear regression tools from the stats, hydrotool, tools and nls packages. Their inputs were limited to experimental infiltration rate and time, allowing direct curve fitting to observed data.

2) Machine Learning Models

Machine learning approaches included a Multilayer Perceptron (MLP) and a Long Short-Term Memory (LSTM) network, implemented in Python (Spyder IDE) with TensorFlow and Keras. Data were split into 70% training, 15% validation, and 15% testing. Inputs combined experimental infiltration rate, soil moisture, soil texture, and time, enabling richer predictive capacity than the empirical models.

3) Data Processing and Training

All data were normalized prior to training. The MLP used dense hidden layers with ReLU activations, while the LSTM included recurrent layers followed by a dense output. Both models were optimized with the Adam algorithm and mean squared error loss, with early stopping applied. Performance

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was evaluated using R^2 and RMSE for direct comparison with empirical results.

III. Results

Fig.2 presents scatterplots comparing observed and simulated infiltration rates for four models: two empirical (SCS and Philips) and two deep learning models (MLP and LSTM). The 1:1 line represents the ideal case where simulated values perfectly match observations.

SCS Model: The SCS model shows moderate agreement with observations but demonstrates overestimation at high infiltration rates (above 400 mm/h). The spread around the 1:1 line increases at higher values, indicating inconsistent performance during peak infiltration.

Philips Model: The Philips model displays clustering around lower observed values and tends to underestimate medium to high infiltration rates. The spread is more dispersed than the SCS model, and the simulation curve diverges significantly from the 1:1 line, especially for values exceeding 300 mm/h.

MLP Model: The MLP model exhibits the strongest alignment with the 1:1 line, indicating excellent prediction accuracy. Both calibration (gray) and validation (orange) points are tightly clustered with minimal scatter, reflecting the model's ability to generalize well across datasets.

LSTM Model: The LSTM model also demonstrates good predictive capability, with most points close to the 1:1 line. However, compared to MLP, it shows slightly greater dispersion, particularly in the mid-to-high infiltration range, suggesting a minor loss in accuracy during validation.

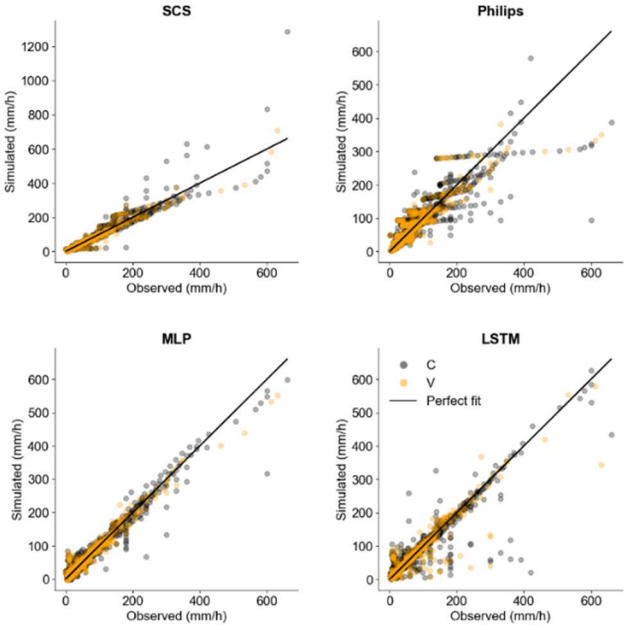


Fig.2. Scatterplot of Comparing Observed and Predicted Infiltration Rates Utilizing Machine Learning and Empirical Models

Fig. 3 summarizes model performance using boxplots for four statistical indicators: PBIAS, correlation coefficient (r), RMSE, and NSE.

A. PBIAS

All models exhibit low median bias, with MLP and LSTM centered close to zero, indicating minimal over- or underestimation. However, empirical models, especially Philips, show more outliers and greater variability, confirming their limited consistency across test cases.

B. Correlation Coefficient (r)

MLP shows the highest median correlation, nearing 1.0, followed closely by SCS and LSTM. Philips underperforms with a lower median and larger interquartile range, confirming weaker predictive relationships. LSTM, while robust, shows a wider spread than MLP, pointing to occasional deviations in certain samples.

C. RMSE

MLP exhibits the lowest RMSE values overall, demonstrating higher accuracy and lower error spread. LSTM follows, with slightly higher error margins but still significantly better than the empirical models. SCS and Philips show wider distributions and higher medians, reflecting their inability to capture the variability in infiltration dynamics effectively.

All models show near-zero or negative NSE values due to extreme outliers. However, MLP consistently maintains closer-to-zero values, while empirical models (especially Philips) show several large negative outliers, indicating poor performance in replicating observed variability. Despite this, the distribution for MLP and LSTM is tighter, signaling better stability and reliability.

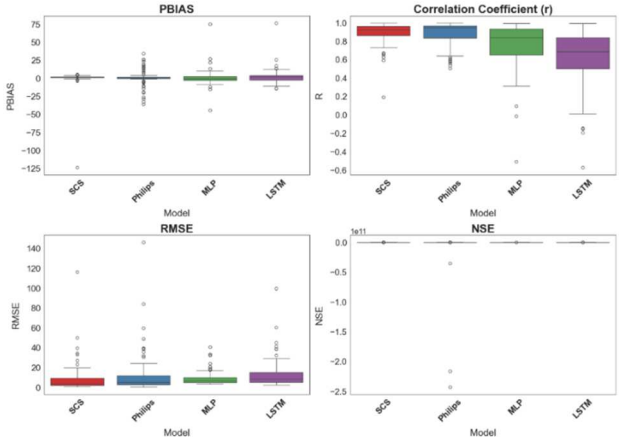


Fig.3. Comparison of ML functions based on NRMSE, RSR, Correlation (r), and PBIAS statistics.

IV. Discussion

The comparative evaluation of empirical and deep learning models revealed clear differences in predictive performance for infiltration modeling in the Mitidja Plain. Among the empirical models, SCS showed relatively stable predictions with moderate correlation but struggled with high infiltration rates, exhibiting a tendency toward overestimation. This aligns with the findings of [7], who reported SCS's limited reliability in accurately predicting infiltration across varying soil textures, particularly under dynamic field conditions.

In contrast, the Philip model exhibited poor performance across all statistical metrics. It underestimated high infiltration rates and produced significant error dispersion, confirming previous criticisms of the model's inability to adapt to diverse infiltration behaviors [15]. Despite its theoretical foundation, Philip's equation often fails to generalize beyond specific soil conditions, especially when

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time-based variability and surface crusting effects are present, as seen in Mitidja's heterogeneous soils.

On the machine learning side, both deep learning models—MLP and LSTM—delivered markedly better performance. The MLP model demonstrated the highest predictive accuracy, with minimal bias and tight clustering around the 1:1 line in both calibration and validation datasets. Its strength lies in its ability to model complex nonlinear interactions between infiltration-driving variables (e.g., initial moisture, texture, and time). These findings support previous work by [16] and [17], where neural networks effectively captured infiltration variability in similar agro-hydrological contexts.

The LSTM model, while slightly more variable than MLP, performed robustly overall. It successfully captured the temporal dynamics of infiltration data recorded at 1-minute intervals, though with marginally wider prediction dispersion. This behavior is consistent with [18] who emphasized that LSTM excels in time-series applications but may require larger datasets and careful regularization to avoid overfitting, particularly in short or noisy series.

These outcomes reflect broader trends in hydrological modeling. Empirical models like SCS and Philip, while computationally simple and interpretable, are increasingly outperformed by data-driven methods in environments with complex, nonlinear infiltration processes. Meanwhile, neural networks—particularly MLP—have shown strong generalization across soil types and conditions, supporting their integration into operational modeling frameworks.

The results underscore the need for hybrid modeling strategies, combining the interpretability and simplicity of empirical models with the precision and adaptability of machine learning. In regions like Mitidja, where soil textures and infiltration responses vary sharply over short distances, such hybrid approaches could provide the most reliable predictions while maintaining transparency for decision-makers.

V. Conclusion

This study assessed the performance of two empirical models (SCS and Philip) and two deep learning models (MLP and LSTM) for predicting soil infiltration rates in the Mitidja Plain. The results confirm that traditional empirical models, while simple to implement, struggle to capture the complex and nonlinear behavior of infiltration processes under variable soil and moisture conditions. In particular, the Philip model showed the weakest performance, while SCS offered moderate accuracy with notable limitations at higher infiltration rates.

In contrast, deep learning models—especially the Multi-Layer Perceptron (MLP)—demonstrated superior predictive accuracy, robustness, and generalization across calibration and validation datasets. The LSTM model also performed well, particularly in capturing temporal dynamics, although with slightly more variability than MLP.

These findings underscore the value of integrating artificial intelligence into hydrological modeling, particularly in heterogeneous semi-arid environments like Mitidja. Machine learning models offer a scalable and accurate alternative to empirical approaches and should be considered for future infiltration modeling and water resource planning. To further improve model reliability and interpretability, future work should explore hybrid frameworks that combine empirical understanding with the flexibility of deep learning.

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